Introduction to Aviation Cybersecurity

CEC 300: Computing in Aviation and Aerospace

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Introduction to Aviation Cybersecurity

- Modern aircraft and aerospace systems blend digital, physical, and network elements.
- Safety and operational reliability depend on robust cybersecurity
- Vulnerabilities exist from sensors onboard to air traffic management centers to airlines to airports.
  - Cyber adversaries target both legacy and state-of-the-art technologies.
  - Integration of heterogeneous systems increases risk.
  - Continuous innovation in defenses is essential.

Brainstorm:
What are some aviation cybersecurity concerns?

What are some potential targets?

What types of cyberattacks could be applied against those targets?

Threats to Aviation Cybersecurity

- Potential targets:
  - Aircraft / aircraft systems
  - Air Traffic Management
  - Airlines
  - Airport Operations
  - Passengers
  - Manufacturers
  - Infrastructure
  - Communication systems
  - Ground components of satellite-based systems

- Approaches
  - Unauthorized access
  - Denial-of-service
  - Man-in-the-middle
  - Data breaches
  - Malware
  - Ransomware
  - Spoofing
  - Social engineering
  - Supply chain attacks

Distribution of Attacks from 2000 to 2024

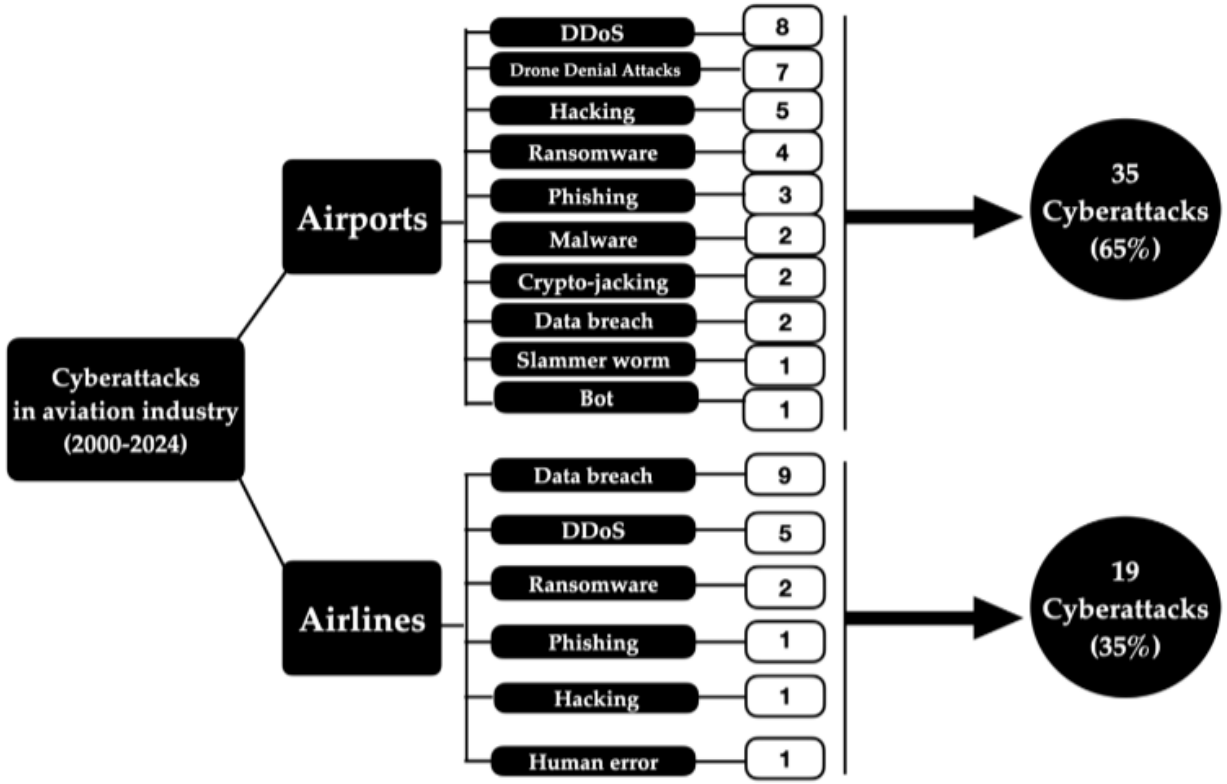
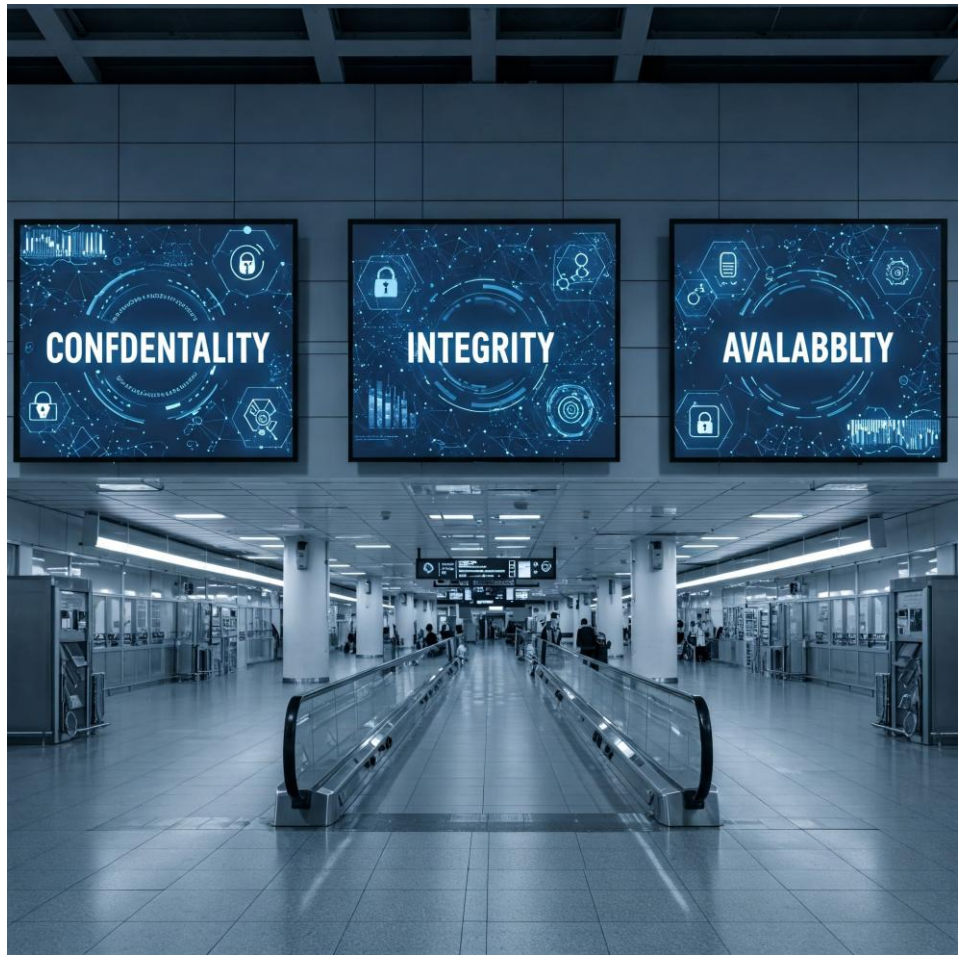


Fig. 4 Cyberattacks perpetrated in aviation industry from 2000 to January 2024. Source. Author's own elaboration from KonBriefing (KonBriefing. 2022) and digital newspapers

Categories of cyberattacks

- **Confidentiality:** Attacks aimed at unauthorized viewing of sensitive information, such as passenger data and flight plans.
- **Integrity:** Attacks focused on improper or malicious modification of data, such as ATM system information or aircraft control systems.
- **Availability:** Attacks intended to disrupt access to critical systems and services, such as denial-of-service attacks on airport networks.



Imagan 3 AI (cannot spell)

Threat Actors and Motivation

- **Terrorist Organizations:** Aim to cause financial losses, create widespread fear, and disrupt critical infrastructure.
- **Cybercriminals:** Seek to instill fear, intimidate, steal, and engage in various forms of criminal activities.
- **Nation-State Actors and Advanced Persistent Threat (APT) Groups:** Often collaborate to steal intellectual property and intelligence to advance their domestic aerospace capabilities, as well as monitor, infiltrate, and subvert other nations' capabilities. Their attacks are often sophisticated and hard to detect.
- **Malicious Insiders:** Airline employees or individuals with knowledge of protocols and vulnerabilities can pose a significant threat.

Impact of Aviation Cyber-Attacks

Safety Compromises: Unauthorized access or manipulation of critical systems can directly impact flight safety and potentially lead to catastrophic events.

Financial Losses: Attacks can cause significant economic damage through operational disruptions, theft of financial data, and damage to reputation.

Data Breaches: Sensitive passenger data and intellectual property can be compromised.

Operational Disruptions: Attacks can lead to flight delays, cancellations, and overall chaos in airport operations.

Reputational Damage: Cyber incidents can erode public trust and damage the reputation of airlines and airports.

Theft of Intellectual Property and Intelligence: State-sponsored actors target the aviation industry to acquire valuable information.

Vulnerabilities Exist Across Systems and Sub Systems. Here are a few to consider.

- Flight Systems:
  - Software-intensive systems like flight control (including autopilot), displays, navigation, communication, engine control, landing gear, and collision avoidance systems can contain vulnerabilities introduced through human error, lack of quality control, malicious code, supply chain of software components.
  - Potential for unauthorized access and tampering with flight control systems.
- Communications:
  - Unencrypted communication protocols, such as ADS-B, are susceptible to spoofing and manipulation and do not maintain confidentiality.
  - Vulnerabilities in Controller Pilot Data Link Communications (CPDLC) could be exploited.
  - Lack of (strong) authentication on critical data flows like data link can be exploited.
  - Weak integrity control on communication data flows allows for potential undetected modifications.
- Surveillance:
  - Spoofing of ADS-B radio signals can present false aircraft information.
  - Spoofing of radar data over interconnection networks can mislead air traffic controllers.
  - Signal jamming of surveillance systems like Primary Surveillance Radar (PSR) and Secondary Surveillance Radar (SSR) can disrupt air traffic monitoring.
- Navigation:
  - GPS spoofing or jamming can provide erroneous navigation information to aircraft.
  - Vulnerabilities in navigation aids like VOR and ILS could be exploited (though the technical difficulty varies).

EXAMPLES OF AVIATION CYBERATTACKS.

Timeline of Cyber Attacks

Year	Events
2003	• The Slammer Worm attack compromised one of the FAA's administrative servers.
2006	• US Federal Aviation Administration (FAA) internet services forced the shutdown of some of its ATC systems
2012	• Canadian infrastructure, including passenger and cargo flights, as well as airport facilities, targeted in acts of terrorism conducted by al-Qaeda.
2013	• A security consultant claimed to have hacked into an aircraft's control system using his PlainSploit Android app. • A malware attack led to the shutdown of the passport control system at the departure terminals of Istanbul Ataturk and Sabiha Gokcen airports, causing flight delays. • Malicious hacking and phishing attacks allegedly carried out by a nation-state targeted about 75 US airports in an attempt to breach US commercial aviation networks.
2015	• In January, a group calling themselves the 'Lizard Squad – Official Cyber Caliphate' claimed responsibility for breaching the website of Malaysia Airlines. • A Distributed Denial-of-Service (DDoS) attack by cyber-criminals affected LOT Polish Airlines' flight-plan IT Network systems at the Warsaw Chopin airport, grounding at least 10 flights.
2016	• More than 60 new ransomware variants appeared. Ransomware attacks targeted entities within the aviation sector, including a UK Airport where screens were blacked out, and Civil Aviation Authorities. • Hacking and phishing attacks led to the defacement of a website belonging to Vietnam Airlines and flight information screens at Ho Chi Minh City and Hanoi.

Timeline of Cyber Attacks (2)

Year	Events
2018	• Boeing was hit by the WannaCry computer virus, although it was reported to have caused minimal damage to the company's internal systems. • A cyber-attack allegedly launched by the Russian APT group (APT28) blocked Sweden's air traffic control capabilities, grounding hundreds of flights over a 5-day period. • A crypto-mining malware was discovered infecting more than 50% of the European airport workstations.
2019	• About 3 million bot attacks were blocked in a single day by Israel's Ben Gurion Airport authority as they attempted to breach airport systems. • A cyber incident resulted in unauthorized access to Airbus' "Commercial Aircraft business" information systems. • Albany International Airport experienced a ransomware attack that encrypted its entire database.
2020	• EasyJet Data Breach: Cyberattack exposed the personal information of approximately 9 million customers.
2021	• SITA Data Breach: Data breach affecting frequent flyer data of multiple airlines. • Air India Data Breach: Data breach compromised the personal data of about 4.5 million passengers.
2022	• Los Angeles Airport DDoS Attack: Website taken offline for several hours due to a DDoS attack.
2024	• Cathay Pacific Breach: Breach affected more than 9 million passengers' personal information.

Unauthorized Access

Unauthorized access refers to gaining access to systems, networks, or data without permission. This can involve exploiting software vulnerabilities, using stolen credentials, or bypassing security measures.

- **Event:** RwandAir Cyberattack
- **Type:** Unauthorized access
- **Actors Involved:** SilitNetwork (suspected)
- **Description:** Hackers gained unauthorized access to RwandAir's systems, exploiting weaknesses in the airline's cyber defenses. Used a combination of social engineering, phishing, and software exploits. Loss of confidentiality and integrity of passenger data. Near loss of critical operational functions.
- **Future Mitigations:** Implementing stronger access controls, regular security audits, and employee training on cybersecurity

Resecurity | The Aviation and Aerospace Sectors Face Skyrocketing Cyber Threats

Denial-of-Service Atack (DOS)

Denial-of-service attacks aim to make systems or networks unavailable to users by overwhelming them with traffic or exploiting vulnerabilities.

- **Event:** Japan Airlines DDoS Attack
- **Type:** Denial-of-service
- **Actors Involved:** Unknown
- **Description:** A DDoS attack on Japan Airlines caused significant flight delays and temporary suspension of ticket sales.
- **Future Mitigations:** Enhancing network infrastructure, implementing DDoS protection services, and conducting regular penetration testing.

Japan Airlines Cyberattack Causes Flight Delays - Security Spotlight

Data Breaches

Data breaches involve unauthorized access to sensitive information, leading to its exposure or theft.

- **Event:** ICAO Data Breach (2025)
- **Type:** Data breaches
- **Actors Involved:** Unknown
- **Description:** The International Civil Aviation Organization experienced a significant data breach, compromising personal information of 11,929 individuals.
- **Future Mitigations:** Strengthening data encryption, implementing multi-factor authentication, and conducting regular security audits.

Resecurity | ICAO and ACAO Breached: Cyberespionage Groups Targeting Aviation Safety Specialists

Malware

Malware refers to malicious software designed to disrupt, damage, or gain unauthorized access to systems.

- **Event:** Continental Aerospace Technologies Cyberattack (2024)
- **Type:** Malware (also a case of ransomware)
- **Actors Involved:** PLAY ransomware group
- **Description:** Continental Aerospace Technologies was targeted by a severe malware attack, disrupting manufacturing operations, and compromising intellectual property.
- **Future Mitigations:** "Robust threat-modeling and the implementation of corresponding measures to protect intellectual property, maintain operational continuity, and ensure the safety of aerospace supply chains."

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Ransomware

Ransomware is a type of malware that encrypts data and demands a ransom for its release.

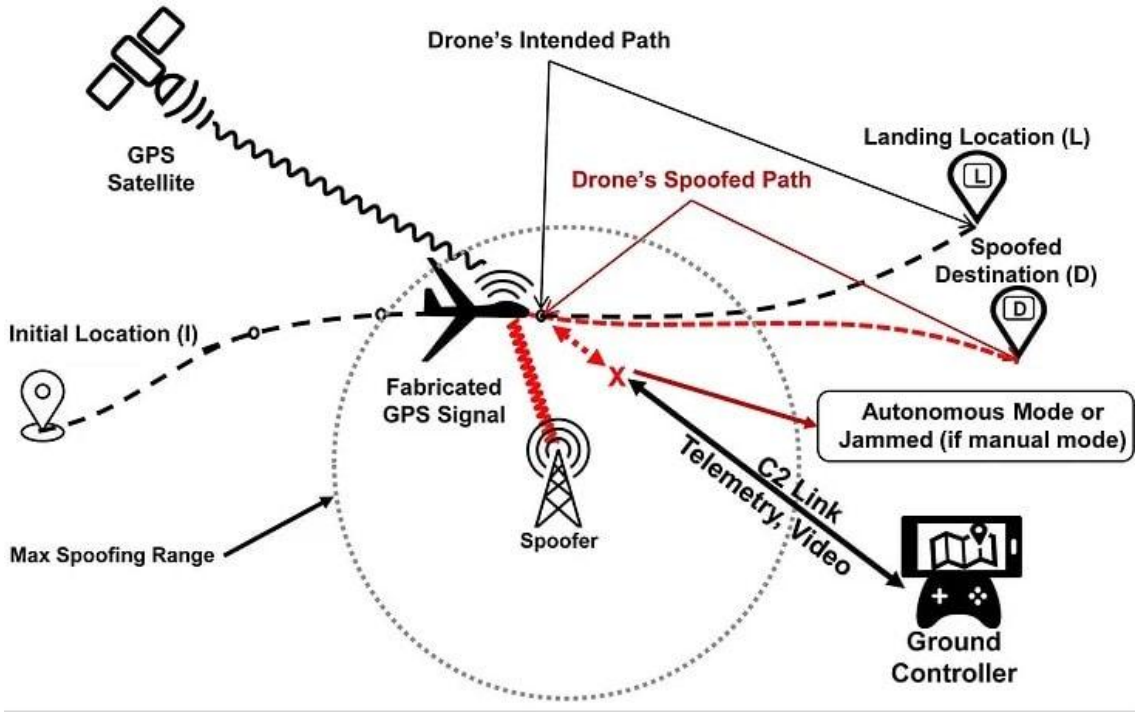
- **Event:** AerCap Ransomware Attack
- **Type:** Ransomware
- **Actors Involved:** Slug
- **Description:** AerCap, a global aviation leasing giant, was hit by ransomware, impacting its IT systems. Over 1TB leak, as claimed by Slug.
- **Future Mitigations:** Regular data backups, implementing strong encryption, and conducting cybersecurity training

AerCap ransomware attack latest to hit aviation sector | Cybernews

Spoofing

Spoofing involves falsifying data or signals to deceive systems or users.

- **Event:** GPS Spoofing in the Middle East
- **Type:** Spoofing
- **Actors Involved:** Unknown
- **Description:** 20+ civilian aircraft were misled by GPS spoofing, resulting flights approaching Iranian airspace without clearance.
- **Future Mitigations:** Industry has adopted various anti-jammer and anti-spoof technologies to mitigate likelihood of disruptive jamming.



[GPS Spoofing: New Electronic Threat in Civilian Airspace - IEEE Spectrum]

Social Engineering

- Social engineering involves manipulating individuals to gain unauthorized access or information.
  - **Event:** British Airways Data Breach
  - **Type:** Social engineering
  - **Actors Involved:** Cybercriminals
  - **Description:** In 2018, British Airways suffered a data breach where attackers used social engineering techniques to trick employees into revealing sensitive information. This led to the theft of personal and financial data of around 380,000 customers.
  - **Future Mitigations:** Implementing comprehensive employee training programs, enhancing phishing detection systems, and conducting regular security audits.

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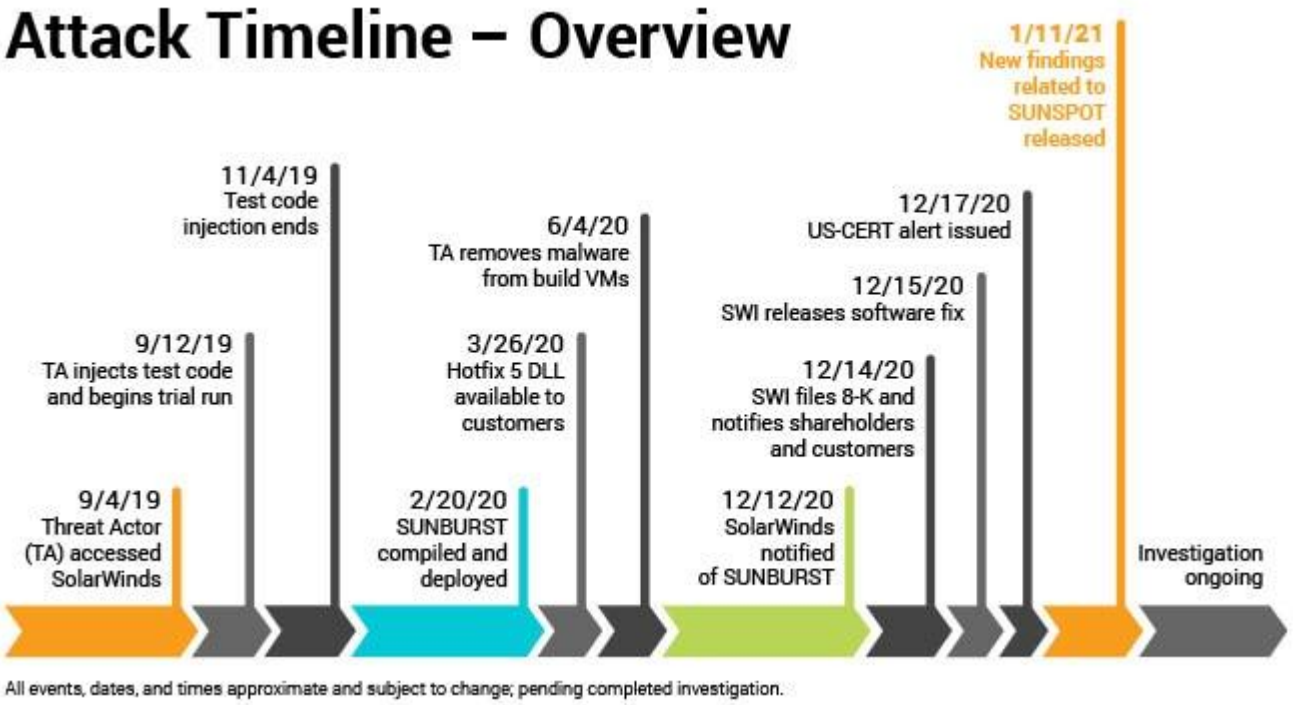
Supply Chain Attacks

Supply chain attacks target vulnerabilities in the supply chain to compromise systems or data.

- **Event:** SolarWinds Attack Impact on Aviation
- **Type:** Supply chain attacks
- **Actors Involved:** Russia's Foreign Intelligence Service (SVR)
- **Description:** The SolarWinds attack in 2020 compromised numerous organizations, including those in the aviation sector. Hackers infiltrated the SolarWinds software update process, allowing them to access sensitive data and systems of various companies, including aviation-related entities.
- **Future Mitigations:** Implementing stringent vetting processes for suppliers, enhancing software update security, and adopting SBOM (Software Bill of Materials) practices.



Note US-CERT: United States Computer Emergency Readiness Team

[Resecurity | The Aviation and Aerospace Sectors Face Skyrocketing Cyber Threats]

SOFTWARE CYBERSECURITY IN AVIATION

Software Challenges – Overview

Challenges:

Software controls flight management, navigation, and communication

Many aircraft systems use custom, legacy code alongside modern software

Complex code integration increases the risk of hidden vulnerabilities

Regulatory delays sometimes impede prompt software updates

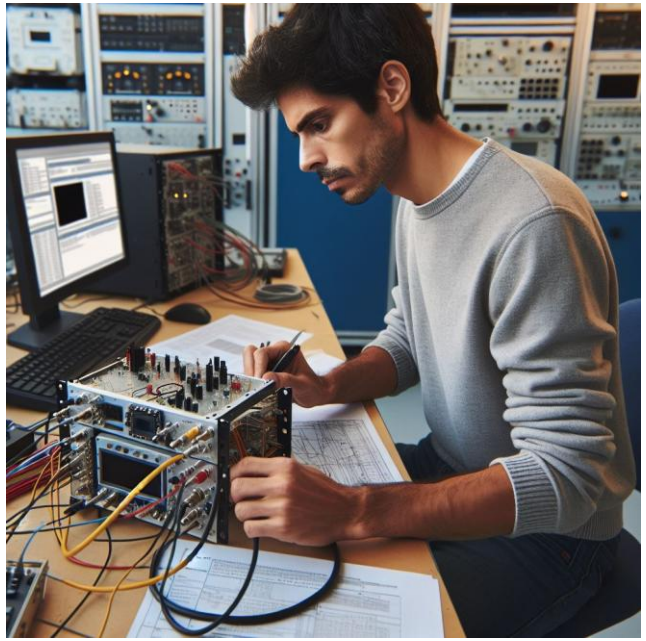
Maintaining secure software under dynamic threat environments is essential

Issue: Legacy Software

- Legacy avionics may have been certified under outdated security standards
  - These systems are prone to exploitation due to unpatched vulnerabilities
  - Upgrading legacy software is challenging and costly in aviation
  - Modernization requires balancing safety, performance, and regulatory compliance
- Older flight control programs often lack modern encryption or authentication

Issue: Firmware Vulnerabilities & Code Injection

- Firmware governs low-level hardware behavior in self-contained avionics
- Unprotected firmware may be exploited via code injection or buffer overflows
  - Example: Attacks against flight computer firmware can compromise control logic Secure boot processes and digital firmware signing mitigate these risks
- Strict code reviews and testing are applied in critical avionics systems
- Regular firmware updates are integral to maintaining operational security



Issue: Insider Threats & Malicious Updates

- Challenges:
  - Insider actions can lead to unauthorized software modifications
  - Compromised update channels risk widespread exploitation in connected systems
- Mitigations:
  - Aviation firms enforce strict access control and audit trails for updates
  - Digital signatures verify the authenticity of software updates
  - Continuous monitoring of update channels is vital
  - Training staff to recognize social engineering increases overall security

Issue: Complexity in Software Integration

- Challenges:
  - Modern aircraft integrate software modules from various vendors
  - Integration issues can expose unforeseen vulnerabilities
- Mitigations:
  - Interfaces between flight management, navigation, and control systems must be rigorously tested
  - Standardized APIs and secure data exchange protocols are adopted
  - Collaborative design practices improve software robustness in avionics
  - Thorough threat modeling ensures compatibility and security

Issue: Resource Constraints in Avionics Software

- Embedded avionics systems have limited CPU and memory resources
  - Lightweight cryptographic functions and minimal IDS are necessary
  - Example: Resource-constrained flight control software balances performance with security
- Mitigations
  - Optimized algorithms are developed specifically for aerospace applications
  - Trade-offs are made to maintain real-time operational performance
  - Efficient software design is essential to protect safety-critical systems

Example: Integrated Modular Avionics

- Defined in RTCA DO-297/EUROCAE ED-124
- **Enables:** availability, integrity, safety, health monitoring and fault management, composability
- **Characteristics:**
  - Module network with common API for SW/HW resource access
  - Simplified integration
  - Shared components between modules means easier maintenance
  - Developers focus on application layer relying upon the
  - Modules can use own data bus or share aircraft data bus such as AFDX

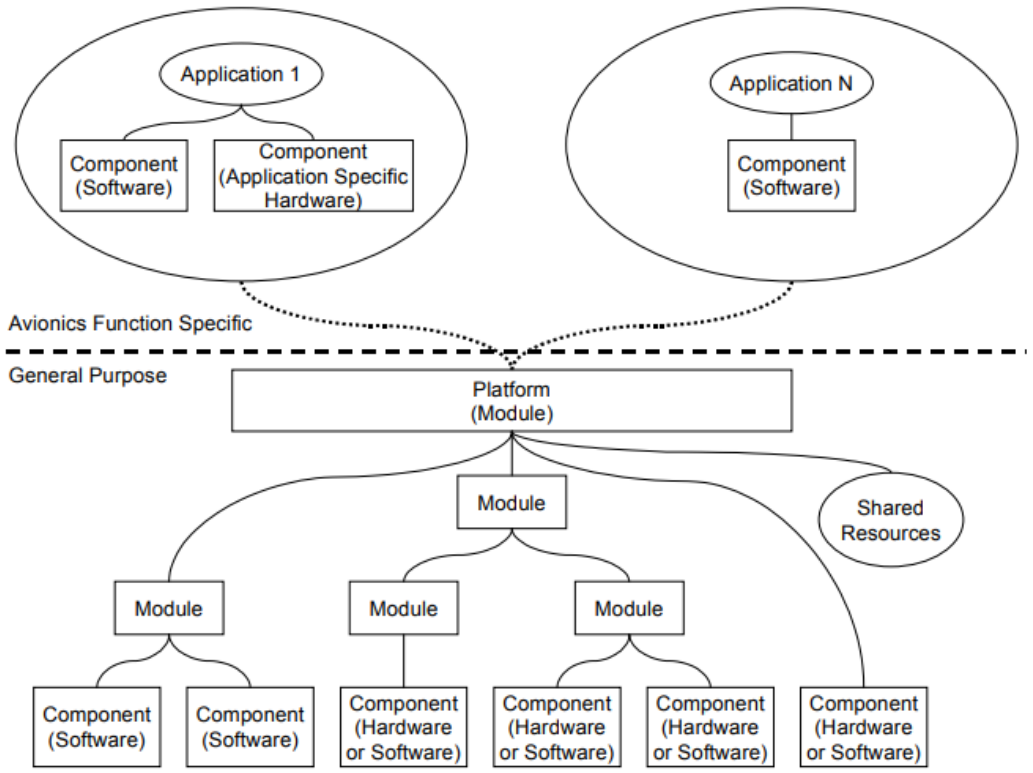
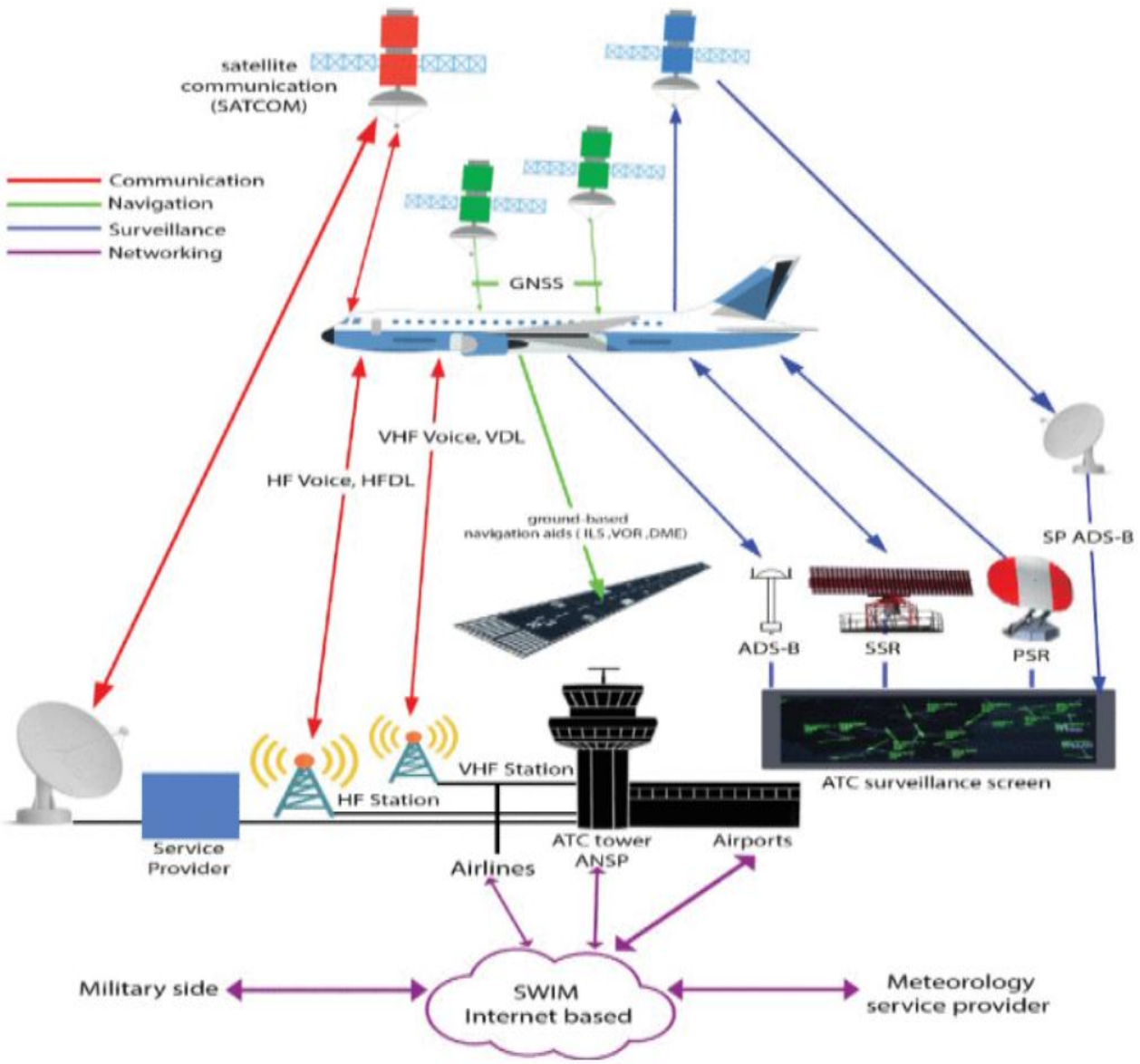


Figure 1: IMA terminology

NETWORKING AND COMMUNICATIONS CYBERSECURITY IN AVIATION

Complex Communication and Networking Environment



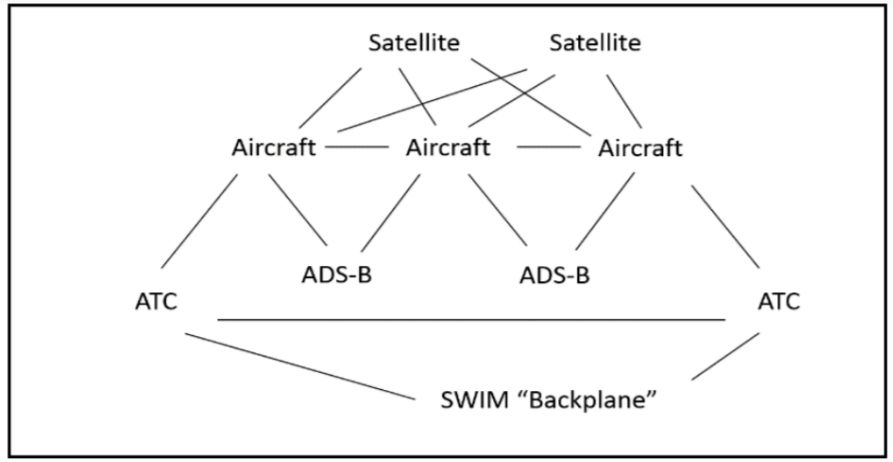


Figure 1. Framework connections and nodes

Cyber Threats to…

Air-Ground Communications

Communication protocol	Threat type / attack scenario
VHF voice	Eavesdropping Jamming (unintentional radio interference due to use unlicensed frequencies) Jamming (unauthorized access to block the VHF frequencies) Spoofing (unauthorized access to send fake instructions between ATC and pilot)
CPDLC	Eavesdropping (listening to the data traffic messages) Jamming (Channel blocking) Flooding (transmitting multiple packets of CPDLC data to the same receiving entity) Injection (sending, possibly faulty, unauthorized messages) Alteration (modification of message content) Masquerading (attacker impersonates an authorized user whether ghost aircraft or ATC)
ACARS	ACARS weight/ balance update and the ACARS flight plan update events

Navigation Aids

Navigation Technique	Threat type
Satellite -based navigation	Unintentional interference Intentional interference Spoofing
ILS	Single-tone attack Overshadow attack
VOR	Jamming Spoofing

Surveillance

Surveillance technique	Threat type
SSR	Jamming Spoofing Over-interrogation
MLAT	Miss synchronization due to vulnerabilities in GPS
ADS-B	Eavesdropping Jamming Spoofing

System Wide Information Management (SWIM)

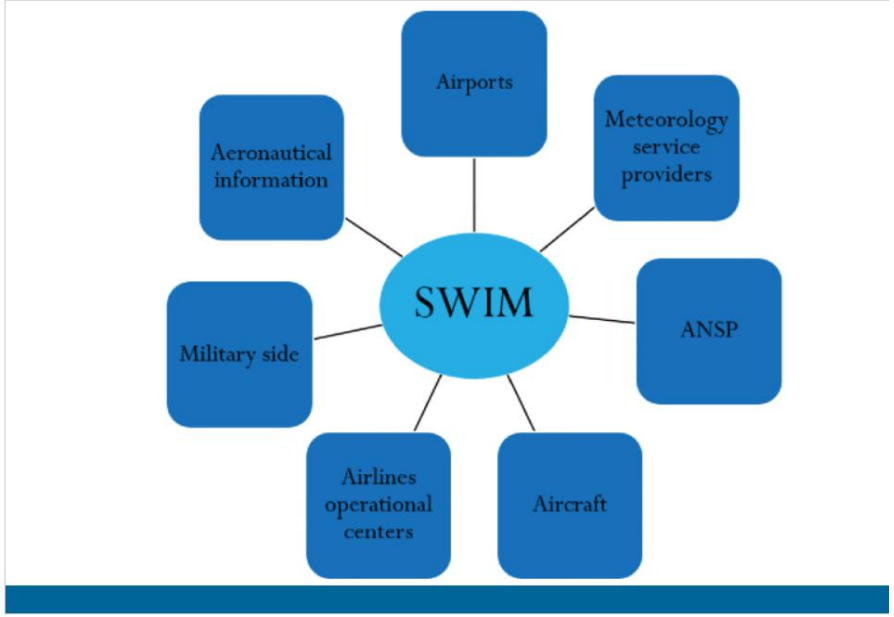


FIGURE 12.
Overview of SWIM and its involved stakeholders.

Potential threats
Man-in-the middle attack
Unauthorized access/DOS attacks
IP-network attacks

SWIM provides a single point of access for near real-time, relevant, and reliable aeronautical, flight, weather, and surveillance information.

- **Infrastructure and Standards:** SWIM delivers the necessary infrastructure, standards, and services to optimize the secure exchange of relevant data across the National Airspace System (NAS) and the aviation community.
- **Data Sharing:** It facilitates the data sharing requirements for NextGen, enabling increased common situational awareness and improved NAS agility.
- **Single Access Point:** SWIM offers a single point of access for aviation data, allowing producers to publish data once and users to access the information they need through a single connection.
- **Operational Excellence:** As the digital data-sharing backbone of NextGen, SWIM supports operational excellence and innovation in aviation.

Issue: Vulnerabilities in Wireless & Satellite Links

Wireless channels are susceptible to interception, jamming, and spoofing

Satellite communication links face interference and cyber manipulation risks

Aviation networks must secure both ground and in-flight communications

Examples include attacks on ADS-B and GPS signals affecting unmanned aerial vehicles

Advanced encryption standards and secure modems are used

Continuous monitoring in ground stations mitigates these risks

Issue: ADS-B & GPS Spoofing Attacks

- ADS-B transmits flight positions without strong authentication
- GPS spoofing can create “ghost” aircraft or falsify navigation data
- Experimental attacks have demonstrated the potential for real-world disruption
- Cross-checking with additional sensors (e.g., radar, inertial navigation) is essential
- Mitigation involves cryptographic signing of messages and anomaly detection
- Lessons learned from avionics testbeds guide secure system designs

TABLE I. SUMMARY OF THREATS, ATTACKS, AND ESTIMATES OF SEVERITY, COMPLEXITY AND EFFECT [6].

Attack	Method	Severity	Complexity	Effect
Aircraft reconnaissance	Eavesdropping	Low	Lowest	Confidentiality
Ground Station Flood Denial	Signal Jamming	Medium	Lower	Availability
Aircraft Flood Denial	Signal Jamming	Medium	Low-Medium	Availability
Ground Station Ghost Injection	Message Injection	High	Low	Integrity
Aircraft Ghost Injection	Message Injection	Medium	Low-Medium	Integrity
Virtual Aircraft Hijacking	Message Modification	High	Medium	Integrity
Virtual Trajectory Modification	Message Modification	High	Medium	Integrity
Aircraft Disappearance	Message Deletion	High	Low	Availability
Aircraft Spoofing	Message Modification	High	Low	Integrity

Issue: Data Injection in Air Traffic Systems

- False data injection can alter flight paths and impact air traffic control
- Unsecured interfaces of air traffic management systems are vulnerable
- Simulated attacks expose the effectiveness of robust data validation methods
- Continuous security audits are applied by aviation authorities
- Immediate isolation of compromised subsystems minimizes potential damage
- Real-time threat detection is critical to maintaining safe operations

Issue: IoT & SCADA in Aviation Systems

- Modern airports use IoT sensors and SCADA systems for monitoring infrastructure
- Insecure IoT devices may be exploited to disrupt airport operations
- Example: Smart sensors in airfield lighting and environmental systems are networked
- Interconnected devices require standard security protocols and access controls
- Advanced IDS, sometimes using ensemble learning techniques, enhance detection Jayalatchumy et al.
- Close coordination between IT and operational groups improves resilience

Ways of Mitigating Communication Vulnerabilities

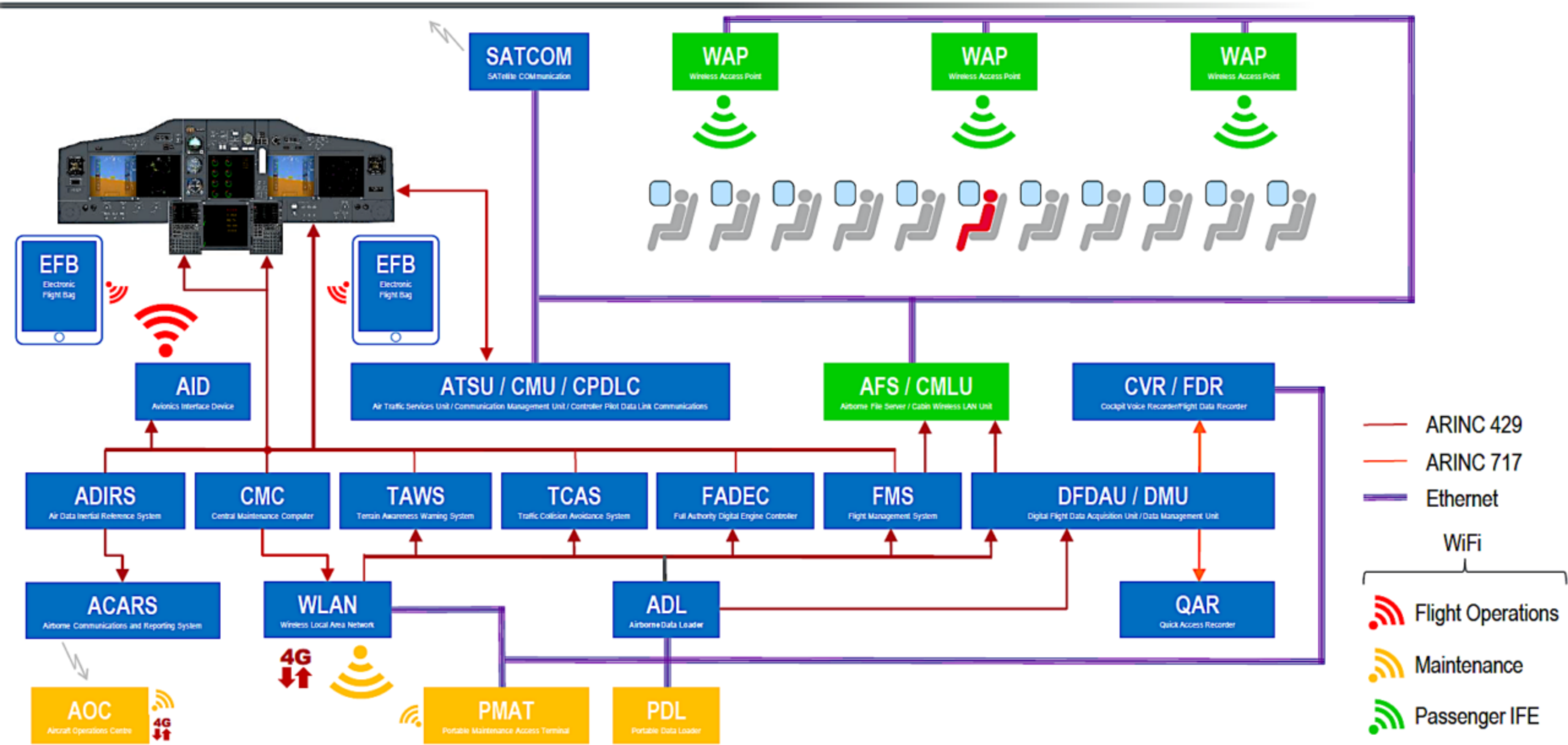
- Apply state-of-the-art encryption (TLS/SSL) to secure data in transit
- Use digital certificates and key exchange protocols (e.g., Diffie–Hellman)
- Implement strict network segmentation to isolate critical aviation channels
- Regular update of communication protocols to patch known vulnerabilities
- Real-time monitoring systems flag anomalies in data transmission

Anomaly detection systems help identify spoofing and injection attacks

- Advanced machine learning algorithms monitor for abnormal data patterns
- Statistical models establish baselines of normal network behavior
- Continuous learning improves detection accuracy over time

HARDWARE CYBERSECURITY

Avionics Cybersecurity Assessment – Passenger IFE



Hardware Challenges – Overview

Critical components: sensors, embedded controllers, actuators in aircraft

Examples: Integrated Modular Avionics (IMA) systems require high integrity

Physical hardware failures can compromise flight safety

Exposure to direct tampering or environmental interference

Strict certification processes in aviation demand high assurance

Hardware security is a foundational component of overall safety

Issue: Embedded Systems Vulnerabilities

AIRCRAFT USE EMBEDDED DEVICES WITH LIMITED COMPUTATIONAL RESOURCES

LEGACY MICROCONTROLLERS MAY LACK ADVANCED SECURITY FEATURES

VULNERABILITIES CAN LEAD TO UNAUTHORIZED FIRMWARE MODIFICATION

RESOURCE CONSTRAINTS LIMIT THE IMPLEMENTATION OF COMPLEX CRYPTOGRAPHY

SECURE DEVELOPMENT PRACTICES MUST ACCOUNT FOR THESE LIMITATIONS

Issue: Sensor & Actuator Integrity

- Sensors (altimeters, gyroscopes) deliver critical flight data
- Actuators (control surfaces, engine controllers) rely on accurate commands
- Incorrect sensor readings may trigger unsafe responses
- Spoofing of sensor outputs can lead to false control inputs
- Regular calibration and redundancy ensure reliability

Issue: Physical Tampering & Side-Channel Risks

- Direct access to hardware (flight computers, wiring) poses serious risks
- Aviation systems often use tamper-resistant enclosures
- Side-channel attacks exploit power consumption or electromagnetic emissions
- Continuous monitoring of physical integrity is recommended
- Preventive measures are essential in high-security areas such as cockpit electronics

Mitigations for Physical Hardware Threats

- **Risks:**
  - Direct access to hardware (flight computers, wiring) poses serious risks
  - Aviation systems often use tamper-resistant enclosures
  - Side-channel attacks exploit power consumption or electromagnetic emissions
  - Continuous monitoring of physical integrity is recommended
- **Mitigations**
  - Hardened Enclosures: Using robust materials like steel or specialized alloys to make it difficult to physically access the components.
  - Security Screws/Locks: Employing screws that require special tools to remove or locks that prevent unauthorized access.
  - Encapsulation/Sealing: Covering components with a sealant or coating that becomes evident if tampered with.
  - Ultrasonic Welding: Joining enclosure parts in a way that is difficult to reverse.

Issue: Counterfeit & Untrusted Components

Supply chains in aerospace may include components from diverse sources

Counterfeit or malicious parts can undermine hardware integrity

Verifying component authenticity is critical for flight-critical systems

- Blockchain traceability can be applied to parts and other system components
- Strict vendor audits and certification procedures are required
- Ensuring trusted hardware mitigates downstream risks

Securing Aviation Manufacturing Facilities

Hardware Mitigation Strategies

- Implement secure boot mechanisms to ensure firmware authenticity
- Use hardware roots of trust and cryptographic modules in avionics
- Apply tamper-resistant coatings, secure enclosures, and physical access controls
- Employ sensor fusion to validate heterogeneous data sources
- Regular diagnostics and integrity verifications maintain long-term security
- Best practices from aerospace certification standards guide implementation

AIR TRAFFIC MANAGEMENT

Importance of ATM Cybersecurity

- Air Traffic Management (ATM) and Air Traffic Control (ATC) systems are fundamental to ensuring the safety of air transportation.
- These systems process critical information essential for maintaining aircraft separation and preventing ground collisions.
- Compromise of these systems can lead to significant safety hazards and potential loss of life.
- Unauthorized actions, including disruption, interception, or manipulation of ATM/ATC information, pose severe operational and security risks.

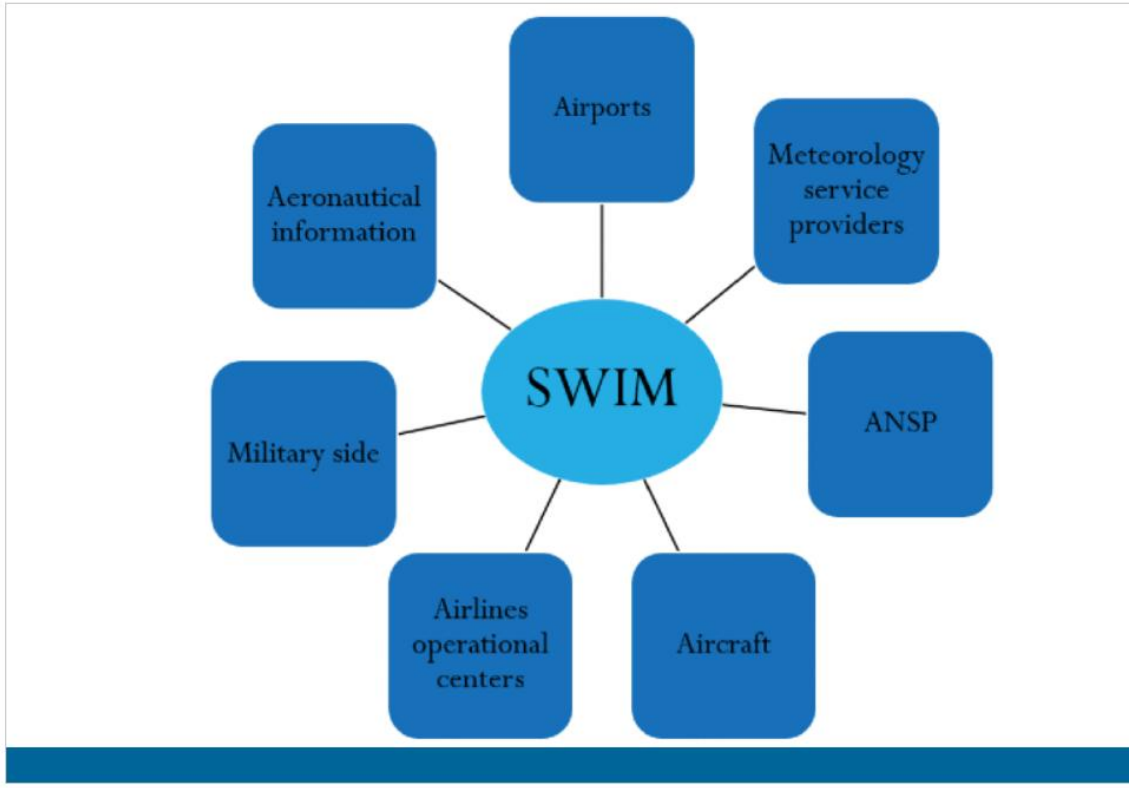


FIGURE 12.
Overview of SWIM and its involved stakeholders.

Cyber Threats Targeting ATM/ATC Systems

- Attacks on **Availability**:
  - **Signal jamming** targeting ADS-B, radar or communication system,
    - Impede communication between controllers and pilots
    - Disrupt traffic flow.
  - **Denial-of-Service (DoS)** or Distributed Denial-of-Service (DDoS)
    - Overload networked or internet connected critical infrastructure, leading to system failures

- Attacks on **Integrity**:
  - Spoofing of ADS-B or radar data
    - False aircraft positions or
    - Conceal actual traffic from air traffic controllers.
  - Disrupt GNSS and/or WAAS signals
  - Injection of false data or commands into communication systems
    - Could lead to incorrect instructions being relayed to pilots or controllers.
  - Manipulation of flight plan data
    - Cause operational confusion
    - Potentially hazardous situations

- Attacks on **Confidentiality**:
  - Mechanisms to obtain private or sensitive information
  - Example: Leaking information regarding owner/operator/passengers on a private flight, military operations, etc.

Mitigation Strategies

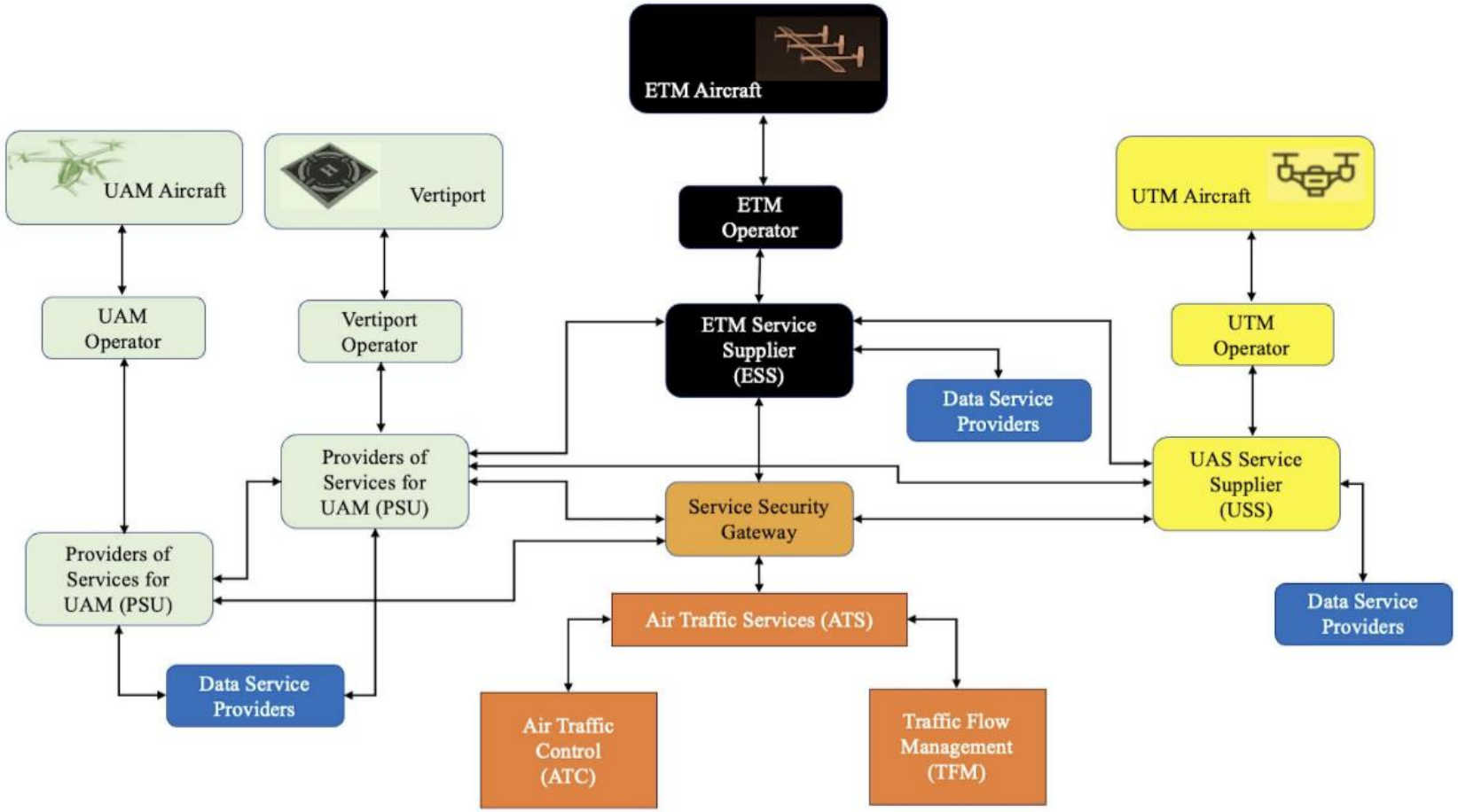
Establishing a process for **continuous vulnerability management**, including regular scanning and remediation of identified weaknesses.

Deployment of **strong authentication mechanisms** for all critical data flows to verify the identity of communicating entities.

Employing **robust integrity controls**, such as cryptographic hashing, to detect any unauthorized data alteration.

Utilizing **Intrusion Detection and Prevention Systems (IDPS)** to monitor network traffic for malicious activity and automatically block detected threats.

UAM and UTM:
Extensible Traffic Management (xTM)



SYSTEMS-LEVEL CYBERSECURITY

System-Level Challenges – Overview

- System-level security covers integration of hardware, software, and communications
- Modern aircraft are cyber-physical systems (CPS) with interdependent components
- Cascading failures in one module can jeopardize entire aircraft operation
- Complex interconnectivity requires holistic risk management
- Coordinated defenses across layers are a necessity

Example: Space Mission Delay Cybersecurity Considerations

Issue: Integration & Interconnectivity Risks

- Multiple avionics subsystems operate in tightly integrated environments.
- Failure in one module may trigger cascading effects across the aircraft.
- Unsecured interfaces between legacy and modern systems are vulnerable.
- Rigorous system architecture reviews and threat modeling are applied.
- Interoperability testing helps reveal integration weaknesses.

Issue: Global Standards & Regulatory Variations

- Aviation operates internationally, with diverse cybersecurity regulations
- Inconsistent policies leave gaps in threat mitigation across borders
- Mitigations:
  - Organizations like ICAO work to harmonize aviation cybersecurity standards
  - Regulatory frameworks guide aircraft manufacturers and operators
  - Global information sharing improves incident response coordination
  - Standardized certification processes drive uniform security measures

Cyber-Physical System Integration Challenges

- Modern aircraft exemplify cyber-physical integration with real-time control systems
  - e.g. Integrated Modular Avionics (IMA) systems combine hardware, software, and networks
  - Fault propagation can occur, impacting multiple interconnected systems
- Recommended solution:
  - Continuous risk assessments and dynamic defense strategies are needed
  - Collaborative engineering practices ensure comprehensive system security

Incident Response & Recovery Challenges

- Timely detection and response to cyber incidents are vital in aviation
- Breaches in air traffic or flight management can disrupt operations dramatically
- Coordinated incident response requires global collaboration among stakeholders
- Real-time monitoring and automated alerts minimize downtime Ismail et al. Detailed recovery plans and regular simulation exercises build resilience
- High-assurance communication with regulatory bodies enhances response efforts

MANAGING CYBERSECURITY

Standards

Organization	Standard Number	Standard Title	Description
RTCA / EUROCAE	DO-326A / ED-202A	Airworthiness Security Process Specification	This document provides a process specification for security aspects of airborne systems throughout their lifecycle. It complements DO-178C and DO-254 by addressing security concerns.
RTCA / EUROCAE	DO-356A / ED-203A	Airworthiness Security Methods and Considerations	This document provides methods and considerations for addressing security within airborne systems. It offers guidance on how to apply the processes defined in DO-326A.
SAE International	AS9100	Quality Management Systems - Requirements for Aviation, Space, and Defense Organizations	While primarily a quality management standard, AS9100 includes requirements related to product safety and security, which implicitly touches upon cybersecurity in connected systems and data management.
SAE International	ARP4754B	Guidelines For Development Of Civil Aircraft and Systems	This aerospace recommended practice provides guidelines for the development of civil aircraft and systems. While not solely focused on cybersecurity, it addresses safety-critical aspects, which increasingly involve security considerations for electronic systems.
SAE International	ARP4761	Guidelines and Methods for Conducting the Safety Assessment Process on Civil Airborne Systems and Equipment	Similar to ARP4754B, this document focuses on safety assessments. Modern safety assessments for complex electronic systems in aerospace must consider potential security threats and their impact on safety.
ISO	ISO/SAE 21434:2021	Road vehicles — Cybersecurity engineering	Although primarily focused on the automotive industry, ISO/SAE 21434 provides a comprehensive framework for cybersecurity engineering throughout the lifecycle of road vehicles. Its principles and processes can offer valuable insights and potentially be adapted for certain aspects of aerospace ground systems and connected aircraft components.
NIST	NIST Cybersecurity Framework (CSF)	Framework for Improving Critical Infrastructure Cybersecurity	While not specific to aviation or aerospace, the NIST CSF is a widely recognized and adopted framework for managing and reducing cybersecurity risks. Aerospace organizations can leverage its principles and best practices to enhance their cybersecurity posture across various domains.
NIST	NIST 800-53	Security and Privacy Controls for Information Systems and Organizations	This publication provides a catalog of security and privacy controls that can be applied to U.S. federal information systems and organizations. Aerospace organizations, particularly those working with government contracts or handling sensitive data, may need to align with or reference these controls.
Cybersecurity and Infrastructure Security Agency (CISA)		Cybersecurity Guidance for the Aviation Sector	CISA provides sector-specific cybersecurity guidance and resources for the aviation industry in the United States. This includes best practices, alerts, and incident response information.

A Framework to Establish Cybersecurity for Aviation Systems

Framework Development – Establish the Foundations

6. Aircraft Cyber Security Program

Establish the plan to mitigate the vulnerabilities identified from the previous activities.

4. Security Log Review

Assess monitoring and detection capabilities

2. Technical Risk Assessment

Technical Security Assessment and / or pentests should be performed during cyber security risk assessment.

5. Training & Awareness

Develop capabilities and asset management foundations to manage and monitor risk assessment findings.

3. Aircraft security handbook

Ensure compliance to security handbooks

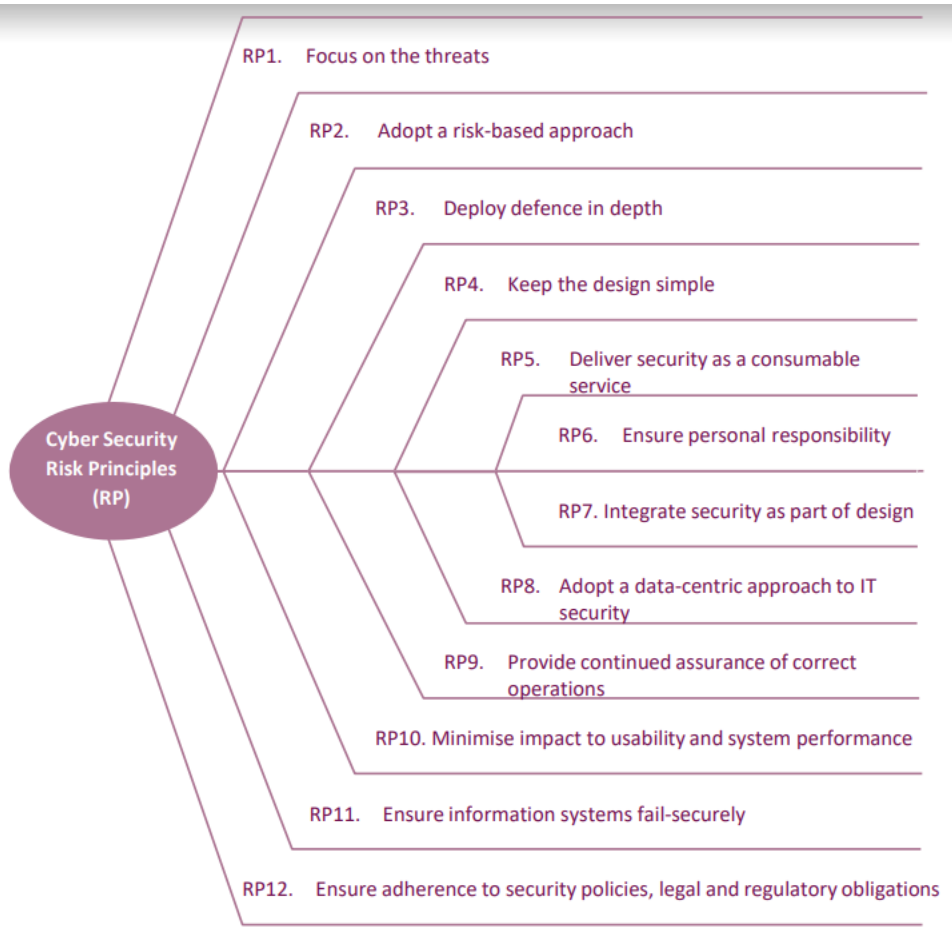
1. Security Risk Assessment

Execute an identification and assessment of risks related to cyber-attacks based on Threat Vectors and Industry standards e.g. NIST (Special Publication 800-30).

Table 1. NIST Core Functions and Objectives. (source – US NIST).

Core Function	Objective
Identify	To understand the cybersecurity environment.
Protect	To develop and implement appropriate safeguards.
Detect	To implement measures to quickly identify cybersecurity events.
Respond	To develop the ability to contain the impact of cybersecurity incidents.
Recover	To develop the capability to restore impaired capabilities and services.

Consideration for Risk Management



RP1: Threats should be understood and documented, including how they are prevented, detected or responded to

RP2: Protection measures should be appropriate to the degree of risk present

RP3: Multiple layers of security should be implemented to mitigate the possible failure of one security component

RP4: The design of an information system must be kept as simple as possible

RP5: Protection measures should be designed to be consumed as common security services

RP6: Users of an information system must be aware of their personal responsibilities to prevent unauthorised access to, and compromise or theft of, information assets and systems

RP7: Security should be treated as an integral part of the overall system design

RP8: Data should be secured end-to-end throughout its lifecycle

RP9: Provide assurance that the information system is, and will continue to be, resilient in the face of (un)expected threats

RP10: The impact of security protections on users and systems should be considered

RP11: In the event of a failure, an information system should fail into a secure state

RP12: The development, build and deployment of systems should be consistent with defined information security policies. legal and regulatory obligations

AI/ML for Cybersecurity

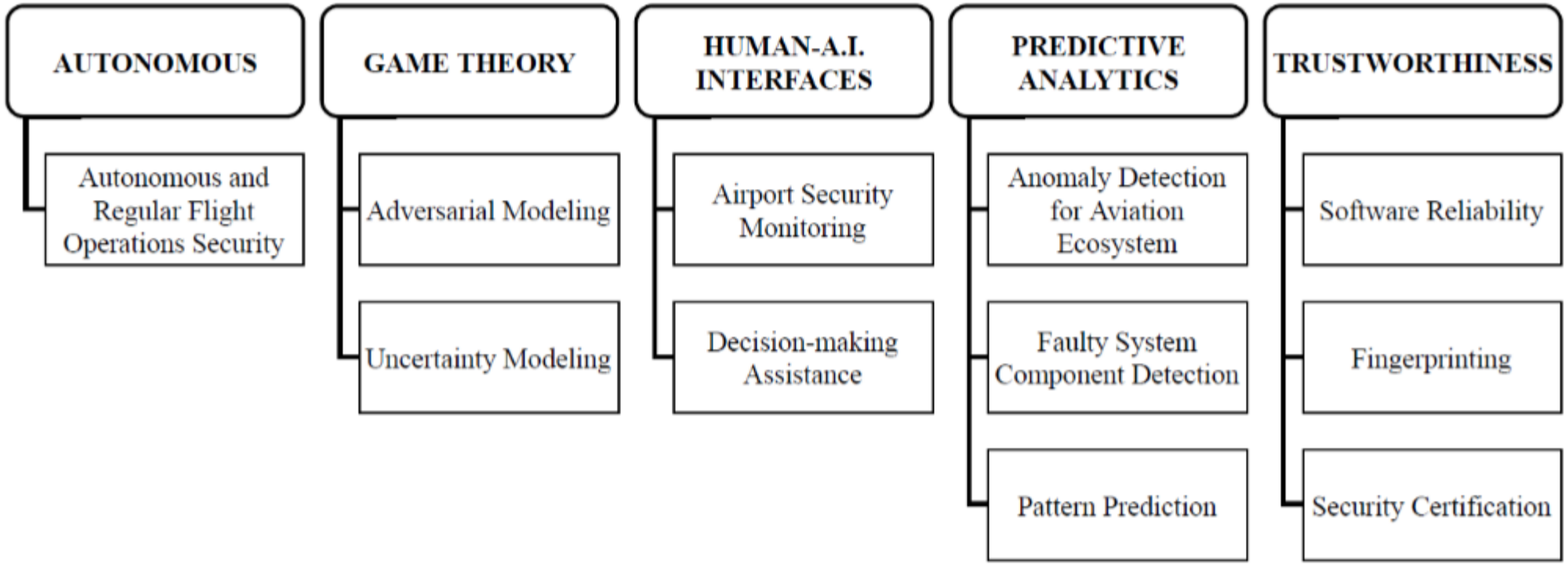


Figure 2. Adapted Types of A.I. and M.L. for Aviation Cybersecurity

Questions